Certified Before Dispatch, Not Better After It: A Pre-Registered Evaluation of a Fail-Closed Arrival-Time Gate for Flood Evacuation

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Abstract

Evacuation decision-support tools usually check whether a road and a shelter are safe *now*, yet a group dispatched now meets the road and the shelter *later*. We built and evaluated a fail-closed gate that dispatches a group only if a one-sided Clopper–Pearson lower bound on the probability that its whole trip succeeds is at least $1 - \varepsilon_{\mathrm{d}}$. Trip success is evaluated jointly within each sampled future: every road segment open when the group enters it, the shelter safe when it arrives, and enough seats left for it after committed reservations and walk-ins. Futures are regenerated from cryptographic keys, every committed decision carries a hash-chained certificate that replays bit-for-bit, and groups on the road are re-certified and migrated atomically when their trip is certified to fail. On a real road network (622 nodes, 1,358 directed edges) with simulated flood hazards, a pre-registered evaluation of 1,260 episodes gave a mixed answer. At the decision level two pre-registered bars passed: the joint indicator blocked fewer feasible dispatches than a product-of-marginals rule (179 vs. 243 false blocks, McNemar $p = 1.08 \times 10^{-19}$) and no sampled state contained a certified dispatch whose oracle success probability was below the threshold (0/400). At the system level the primary bar failed: the gate cut the group failure rate from $23.7\,\%$ to $0.7\,\%$ only by dispatching far fewer groups, delivering $21.4\,\%$ of the people a lookahead planner delivered (pre-registered guard: 95 %). Keyed common random numbers did not reduce variance. We therefore make no claim that the gate improves evacuation outcomes, and we discuss what a per-decision safety certificate does and does not buy a deployed system.

1 Introduction

Flood evacuation in dense coastal towns is a sequence of small, consequential decisions: which waiting group to send, to which shelter, by which road, and when. Decision-support systems in this space typically filter actions with a mask evaluated at decision time—a road is usable if it is open now, a shelter is admissible if its current hazard is below a threshold and it has free seats now. The failure mode is familiar to practitioners: a route chosen at the start of a trip can become unsafe during it (Authors to verify 2023), and seats free at dispatch can be taken by walk-ins before the group arrives.

We asked a narrower question than "can AI plan evacuations better?": *can a dispatch be held to a statistical safety guarantee about what happens on arrival, and does holding every dispatch to that guarantee make evacuations better?* We built a gate that answers the first part with a certificate and evaluated the second part under a protocol written and hash-pinned before any evaluation run. The gate has four ingredients:

1. **Keyed futures.** Possible hazard futures are regenerated from a root key derived from the current observation and an index k; every random draw is keyed by (root, k, stream, absolute time, entity), never by draw order or by which candidates are being evaluated.
2. **A joint arrival-time indicator.** For a candidate dispatch and one future, the group is followed along its path to its own arrival time; success requires every segment open when entered, the shelter safe at arrival, and seats free at arrival—all in the *same* future.
3. **A sequential fail-closed certifier.** Candidates are certified with Clopper–Pearson bounds at a Bonferroni-split level; candidates that are neither proven feasible nor proven infeasible are blocked.
4. **An actuation gate with replay.** A dispatch is committed to the capacity ledger together with a hash-chained certificate in one transaction; any third party can regenerate the decision from the certificate's stored inputs. Groups in transit are re-certified with a looser keep threshold and migrated atomically when their remaining trip is certified to fail.

Our contributions are (i) the design and an end-to-end implementation of this gate inside an operator-facing console with a simulated hardware-in-the-loop rig; (ii) a pre-registered evaluation with four pass/fail bars, seven systems, three hazard regimes and an oracle study over 400 decision states; and (iii) an honest report of its result. Two decision-level bars passed and the primary outcome bar failed: the certified gate made no false accept in the sampled states and the joint indicator is less wasteful than marginal rules, but the gate as configured is too conservative to improve outcomes. We think this negative result is the useful part for a

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deployable-AI audience: a per-decision guarantee can be real and still be the wrong objective for the system.

2 Related Work

Evacuation planning and shelter allocation. Optimisation models for evacuation are surveyed by Hamacher and Tjandra (2002). Capacity-constrained routing reserves node and edge capacity at expected arrival times (Lu, George, and Shekhar 2005); multi-period and stochastic models allocate shelters under time-varying hazards, road disruption and shelter loss (Gama, Santos, and Scaparra 2016; Bayram and Yaman 2018), and chance constraints on shelter utilisation appear in risk-averse location models (Authors to verify 2019). These are planning models solved before or between events; they do not gate individual online dispatches with a per-decision guarantee. Real-time rerouting after incidents (Lim, Baharnemati, and Kim 2016) and online shelter replanning for vehicles en route (Idoudi et al. 2023) address in-transit change but without certification. Patented systems predict until when each path segment stays safe and regenerate routes (Corners Co. 2025), filter shelters whose roads are inaccessible or that are at capacity now (Accenture 2022), or predict evacuation time with stochastic Petri nets (NEC 2022); none couples path, shelter state and capacity at the sampled arrival time within one future or gates actuation on a confidence bound.

Safe decision-making. Shielding blocks unsafe actions of a learned policy (Alshiekh et al. 2018); see García and Fernández (2015) for a survey of safe reinforcement learning. The scenario approach turns chance constraints into sampled constraints with a priori violation bounds (Calafiore and Campi 2006), and has been combined with control barrier functions (Do Nascimento, Papachristodoulou, and Margellos 2024). Rollout-based ambulance dispatch samples scenarios at each decision (Authors to verify 2024), but with independent scenarios per decision and no certificate. Our gate borrows the sample-and-bound idea, applies it to a joint arrival-time event over a shared capacity ledger, and makes the undecided case fail closed.

Common random numbers and replay. Counter-based generators make draws a pure function of a counter and a key (Salmon et al. 2011); Vivarium keys draws by decision point and clock so simulated scenarios stay comparable when the number of draws changes (Institute for Health Metrics and Evaluation). Common random numbers do not always reduce variance (Wright and Ramsay 1979)—our Bar 3 is a case in point. Hash chains for tamper-evident records go back to Haber and Stornetta (1991); our certificates chain decisions and additionally store what is needed to regenerate the futures behind them.

3 Setting

World. The road network is a compact extract of a real coastal town from OpenStreetMap (OpenStreetMap contributors) (622 nodes, 1,358 directed edges) with 4 waiting zones and 5 shelters. Hazards are simulated: an AR(1) regional shock plus local noise drives zone and shelter severities; edges close with a probability that grows with hazard and stay closed; vehicles slow down on flooded edges; walk-ins arrive at shelters at a rate that grows with zone hazard and take seats without reservations. A decision epoch is 2 simulated minutes; a trip spans several epochs, so a group can be *stranded* (an edge on its path closes before it is entered), arrive at an *unsafe* shelter, or find it *overflowing*. All randomness is drawn from keyed streams, so every system sees the same worlds for the same seed. Episodes average 312.1 people in the primary regimes. Three regimes are used: *medium* and *high* hazard (primary) and a *stress* regime with a higher walk-in rate (secondary).

Actions and the existing mask. Each epoch the decision layer may dispatch one waiting group from a zone to a shelter along the current shortest open path, or wait (HOLD). The baseline mask allows a dispatch if the destination's severity is below 0.70 at decision time. With $|Z| \cdot |S|$ dispatch actions plus HOLD the fixed action budget is $A_{\max} = 21$.

4 Method

Keyed futures. At each epoch the gate takes a *forecast snapshot*: observed zone and shelter severities and trends, shelter statuses and current edge states—never the simulator's latent state. Its root key is $\rho = \mathrm{SHA256}(\text{incident}, \text{forecast hash}, \text{graph fingerprint}, \text{ledger hash}, \text{params})$. Future k re-runs the simulator's own hazard, closure, degradation and walk-in dynamics from the snapshot, with every draw computed as $\mathrm{SHA256}(\rho, k, \text{stream}, \text{key}, \text{lane})$, where the key holds absolute time and entity identifiers. Because no draw depends on candidate identity, order or count, the certified set is invariant to adding, removing or reordering other candidates, and candidates are compared on common futures. A test confirms that two simulators differing only in latent state produce identical futures.

Joint arrival-time indicator. For candidate c (zone, shelter, people, path) and future f, the rollout walks the path from the departure minute plus boarding time, checking each edge is open in f at its entry step and accumulating travel time (flood slow-down and the keyed travel noise the live simulator uses). At the arrival step a it checks the shelter is not closed and its severity is below $\theta = 0.70$, and that

$$\text{cap}_s - \text{occ}_s - \textstyle\sum_{\text{pending}\to s} \text{people} - \sum_{t \le a} \text{walk-ins}_s(t) \;\ge\; \text{people}_c + m_s,$$

with all committed reservations counted regardless of their arrival time and m_s a sensor-uncertainty margin (zero in simulation). The indicator $z(c, f)$ is 1 only if all three hold *in the same future*; a utility (people delivered, or a weighted loss for each failure kind) is recorded alongside. Optional document rules (below) add further predicates at arrival.

Sequential fail-closed certification. The certifier runs looks at $n = 256, 512, 1024$ futures (the evaluated configuration). After each look it computes one-sided Clopper–Pearson bounds (Clopper and Pearson 1934) on $P(z = 1)$ at level $\alpha = \delta/(L \cdot A_{\max})$, with L looks and the *fixed* budget $A_{\max}$ rather than the number of surviving candidates, so that the error split cannot be gamed by the candidate set. A candidate is FEASIBLE if its lower bound is $\ge 1 - \varepsilon_{\mathrm{d}}$,

INFEASIBLE if its upper bound is $< 1 - \varepsilon_{\mathrm{d}}$, and otherwise UNDECIDED—and UNDECIDED is *blocked*. We use $\varepsilon_{\mathrm{d}} = 0.05$ and $\delta = 0.01$. Among feasible candidates the one with the highest mean utility over the common futures is chosen; if none is feasible the gate holds and escalates to the operator. In a synthetic Bernoulli check of a single candidate (default parameters: up to four looks and 2,048 futures; 10,000 repetitions per level) the acceptance rate was $0.0\,\%$ at true $p = 0.94$, $5.7\,\%$ at $p = 0.96$ and $100.0\,\%$ at $p = 0.99$: the gate essentially never accepts a candidate below the threshold, and rarely one just above it.

Gate, certificate and replay. The chosen dispatch is reserved in the capacity ledger and a certificate is appended in one all-or-nothing transaction; only then are vehicle and shelter-admission commands emitted, each naming its certificate. The certificate holds the root key, the forecast, graph and ledger hashes, the parameters, per-candidate verdicts, bounds and outcome counts, the operator action and the hash of the previous certificate; floats are rounded before hashing so replay is portable. Stored inputs (forecast snapshot, ledger view, candidates) let anyone regenerate the futures and re-run the certifier. Actuators reject commands whose certificate does not verify.

In-transit re-certification and migration. At every new observation each group on the road is re-certified for its *remaining* trip, excluding its own reservation, at the looser keep threshold $1 - \varepsilon_{\mathrm{k}}$ with $\varepsilon_{\mathrm{k}} = 0.10$ (hysteresis). Only a certified failure (INFEASIBLE) triggers a migration: alternatives from the group's next node are certified at $1-\varepsilon_{\mathrm{d}}$, and the old reservation is released and the new one acquired in one transaction with a MIGRATION certificate. A cooldown of 3 epochs, at most 2 migrations per group and a ban on returning to a shelter already left prevent oscillation.

Tiered document rules. Rules drawn from national flood guidelines (National Disaster Management Authority, Government of India 2008) (e.g., a maximum destination severity) can enter the joint indicator as arrival-time predicates, but only at the trust tier their evidence earns: T0 if the stored source passage or verbatim span no longer verifies (a stricter fallback applies and is audited), T1 if provenance verifies, and T2 if additionally an LLM claim-support verifier (OpenAI 2025) judges the rule fully supported by the cited sentences. That verifier is used for T2 only because it passed its own frozen evaluation on 135 constructed claim–evidence pairs (T2 precision 1.000, unsafe acceptance 0.000, T2 recall 1.000, macro-F1 0.970); an earlier configuration failed (macro-F1 0.585, span-integrity failure 0.104) and stays on record as failed. In the live tiering run both approved rules stayed at T1 (NDMA-R01: T1; NDMA-R02: T1): the verifier judged them only partially supported, correctly, because each adds a numeric threshold the guideline sentences do not state. The tier changes provenance labels, not parameters. The outcome evaluation below ran without document rules.

5 System

The gate runs inside a five-page operator console (web front end over a REST/WebSocket API). In a live certified session the operator sees the pending certification, the reasons behind it and comparisons with alternatives, and can ACCEPT, REJECT, or OVERRIDE to another *certified* alternative with a reason; an override to an uncertified action is refused and logged, and nothing is committed. A guided demonstration replays a recorded run of the real engine that shows a HOLD (all options undecided), normal dispatches, an in-transit migration and a cooldown, and any of its certificates can be replayed (match) or tampered with (mismatch) from the console. A hardware-in-the-loop rig (single-board computer, microcontroller nodes for water level and shelter people-counting, a vehicle terminal and an admission relay that verify certificates before acting) is implemented against a simulated publisher; no bench measurements exist yet.

6 Pre-Registered Evaluation

The protocol was written before any evaluation run and pinned by its SHA-256 in an acceptance test; any change after the first run would have required a dated amendment (none was made). The pre-registration is internal—committed to a version-controlled repository with timestamps—not lodged with a public registry. One analysis script was run once; it was later re-run and reproduced byte-identically.

Systems. **A**: greedy risk-weighted planner with the decision-time mask (the existing decision layer). **A2**: two-step lookahead planner with the same mask (reference for people delivered). **HOLD**: always wait (lower anchor). **B**: a combination of known techniques: deterministic "safe-until" edge disabling at hazard ≥ 0.70, a current-capacity shelter filter, and independent Monte Carlo per candidate (128 futures, accept if mean joint success ≥ 0.95; no bound, no common futures). **P**: the certified gate with re-certification. **P-noFC**: P but UNDECIDED candidates with point estimate ≥ 0.95 are allowed. **P-noHyst**: P with $\varepsilon_{\mathrm{k}} = \varepsilon_{\mathrm{d}}$ and no cooldown. Episodes run without a human (automatic acceptance).

Design. 60 unused seeds per regime, every system on the same worlds: 7 systems $\times$ 3 regimes $\times$ 60 seeds $= 1{,}260$ episodes, all completed (a budget rule fixed in advance would have halved the seeds if the projected run time exceeded 48 h; the projection was 30.6 h).

Questions and bars. **Q1 (outcomes).** Primary endpoint E1: group failure rate (stranded + unsafe + overflow groups over dispatched groups) per episode, paired difference P$-$A per (regime, seed), 95 % paired bootstrap CI (10,000 resamples), primary regimes pooled. Guard: R = mean people safely arrived under P divided by that under A2. *Bar 1 passes iff the CI upper bound is below 0 and* $R \geq 0.95$. **Q2 (coupling).** From the P episodes, every 5th decision state (at most 400) is stored replayably; an oracle estimates each candidate's joint success probability p^* on 4,096 disjoint futures; candidates with $|p^* - 0.95| < 0.007$ are excluded as borderline. On the first 1,024 futures we compare the joint certifier with a product-of-marginals rule (Clopper–Pearson

Bar	Pre-registered test	Result	Pass
1 (outcomes)	E1 CI upper < 0 and $R \geq 0.95$	E1 $= -0.230$ $[-0.264, -0.197]$; $R = 0.214$	**no**
2 (coupling)	joint has fewer errors than each marginal rule, McNemar $p < 0.05$	errors 179 vs 243 / 243; $p = 1.08 \times 10^{-19}$	yes
3 (keyed CRN)	median $\text{Var}_{\text{indep}}/\text{Var}_{\text{CRN}} \geq 2$	1.071 (50 states)	**no**
4 (safety)	share of states with a false accept ≤ 0.01	0/400 (CP 95 % upper 0.0075)	yes

Table 1: Pre-registered bars. Bars 2 and 4 pass; Bars 1 and 3 fail. E1 is P−A in group failure rate (120 paired episodes); R is people safely arrived under P relative to A2.

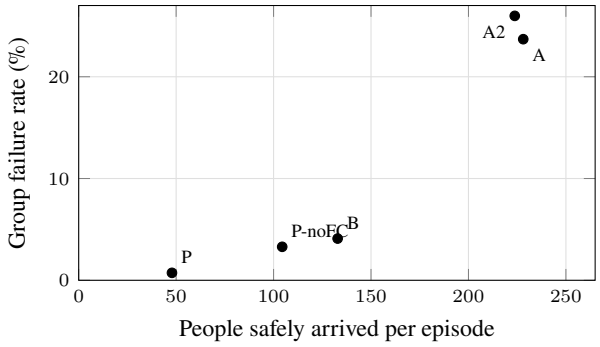


Figure 1: The trade-off behind the Bar 1 failure (primary regimes). P has the lowest failure rate and delivers the fewest people of any dispatching system. P-noHyst coincides with P and is omitted; HOLD sits at the origin.

System	Groups / ep.	Safe arrivals / ep.	Group failure rate
HOLD (always wait)	0.0	0.0	
A: greedy + decision-time mask	16.6	228.1	23.7 %
A2: lookahead-2 + mask	16.6	223.8	26.0 %
B: deterministic + independent MC	8.2	132.9	4.1 %
P: certified gate (ours)	3.3	47.9	0.7 %
P-noFC: no fail-closed	6.6	104.4	3.6 %
P-noHyst: no hysteresis	3.3	47.9	0.7 %

Table 2: Means per episode over the two primary regimes (120 episodes per system). Failure rate is the per-episode group failure rate averaged over episodes. Per-regime numbers, including the stress regime, are in Appendix Table 3.

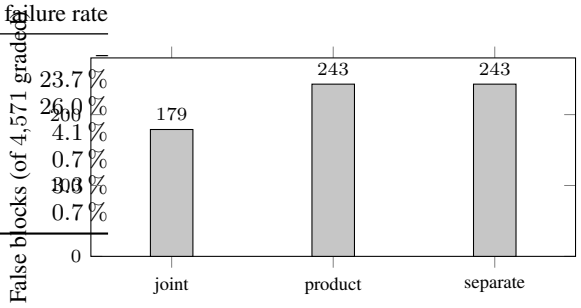


Figure 2: Q2: false blocks among 4,571 graded candidates (4,798 sampled, 227 borderline excluded). No rule made a false accept.

bound on $p_{\text{path}}\, p_{\text{shelter}}\, p_{\text{capacity}}$ from disjoint future blocks) and with separate checks (each marginal's own bound ≥ 0.95). An error is a false block ($p^* \geq 0.95$ but not FEASIBLE) or a false accept (FEASIBLE but $p^* < 0.95$). *Bar 2 passes iff the joint rule has fewer errors than each marginal rule with a two-sided exact McNemar test* (McNemar 1947) $p < 0.05$. **Q3 (keyed CRN).** For the top two candidates in 50 states, the variance of the paired utility difference with common versus independent futures (200 repeats). *Bar 3 passes iff the median variance ratio is* ≥ 2. **Q4 (safety).** *Bar 4 passes iff the share of states containing a FEASIBLE candidate with* $p^* < 0.95 - 0.007$ *is* ≤ 0.01. **Q5 (latency, secondary, no bar).** Targets: certify-only p95 ≤ 100 ms and end-to-end p95 ≤ 10 s at $n = 256$. The decision rule ("D-27") was fixed in advance: if Bar 1 fails, no outcome-effect claim is made.

7 Results

Q1: no outcome effect (Bar 1 fails). The certified gate lowered the group failure rate from 23.7 % (A) to 0.7 % (P); E1 $= -0.230$, 95 % CI $[-0.264, -0.197]$. It did so by dispatching 3.3 groups per episode against 16.6 for A, and it delivered 47.9 people per episode against 223.8 for A2, so $R = 0.214$, far below the 0.95 guard (Table 2, Figure 1). Under the pre-registered rule we make **no claim that the gate improves evacuation outcomes**: not better outcomes, not fewer failed trips at equal throughput. Failure rates are reported only next to R.

Secondary contrasts (reported, no bar). Against B, P's failure rate was lower by -0.034 (CI $[-0.049, -0.019]$), but B delivered 132.9 people per episode against P's 47.9. P-noFC, which lets through undecided candidates with a high point estimate, dispatched twice as many groups as P (6.6 vs. 3.3) and delivered 104.4 people at a failure rate of 3.3 %. P-noHyst was identical to P in every episode: P migrated only 2 times in the primary regimes, so hysteresis never bound and the ablation is uninformative. For executed P dispatches, the median absolute difference between the arrival step the rollout predicted in the true future and the live arrival step was 0.

Q2: joint coupling helps at the decision level (Bar 2 passes). The joint certifier made 179 errors against 243 for the product of marginals and 243 for separate checks; all errors were false blocks, with 0, 0 and 0 false accepts respectively (Figure 2). The joint rule was right on 64 candidates where the product rule was wrong, and the reverse happened on 0 ($p = 1.08 \times 10^{-19}$; 64 vs. 0 against separate checks). The false-block reduction against the product of marginals was 26.3 %, short of a 30 % secondary target.

Q3: keyed common futures did not reduce variance (Bar 3 fails). The median variance ratio was 1.071 over 50 states, far from the $2\times$ bar. Keyed futures remain useful for order invariance and replay, but we claim no variance-reduction effect.

Q4: no false accepts in the sampled states (Bar 4 passes). 0 of 400 states contained a false accept (Clopper–Pearson 95 % upper bound 0.0075).

Supporting checks. Replay: $1{,}274$ of $1{,}274$ certified decisions from 64 episodes were regenerated bit-for-bit from their stored inputs (a fast test parameter set, $\varepsilon_{\mathrm{d}} = 0.30$). Oscillation: in a synthetic study ($1{,}000$ seeds $\times$ 40 steps, success probabilities following random walks with random collapses, not the simulator), 854 migrations in 671 seeds produced 0 returns to a shelter already left and never more than 2 migrations per group. Latency (Q5, laptop, $n = 256$): certify-only p50 288 ms, p95 $1{,}059$ ms; end-to-end p50 6.6 s, p95 13.9 s. Both targets were missed; the single-board computer and energy were not measured.

8 Discussion: Why a Certified Gate Did Not Help

The two halves of the result are consistent. Bar 4 says that what the gate certifies has, in the sampled states, an oracle success probability at or near the threshold or above it; Bar 1 says that holding *every* dispatch to that standard costs most of the throughput. The rest of this section is interpretation, not pre-registered analysis.

The guarantee is per dispatch; the objective is per person. E1 is conditional on being dispatched. A gate that holds people in a flooding zone scores well on E1 while the people it does not move carry the risk; HOLD dispatches no one and so never fails a trip. A gate deployed for its per-decision guarantee needs a system-level objective—people safely delivered, time at risk—as its acceptance criterion, which is why the guard R was in the protocol.

Fail-closed plus multiplicity is conservative by construction. With δ split over $L \cdot A_{\max}$ comparisons and undecided candidates blocked, a candidate needs a true success probability well above $1 - \varepsilon_{\mathrm{d}}$ to be certified within 1,024 futures; in the Bernoulli check, even with up to 2,048 futures, a candidate at $p = 0.96$ was accepted only $5.7\,\%$ of the time. The P-noFC numbers are consistent with fail-closed blocking being one large part of the throughput loss, but P-noFC was a secondary system, not a tested hypothesis.

The joint indicator is the part that travelled. Evaluating path, shelter and capacity in the same future rejected fewer feasible dispatches than treating them independently, at no observed safety cost. Hazards that close roads and degrade shelters share drivers, so marginals misstate the joint event; this matters for any sample-based safety check over coupled resources, independently of whether the check is used as a hard gate.

What would have to change. A gate that should improve outcomes would need its thresholds chosen against a system-level objective, a different response to UNDECIDED than "block" (for example, escalation to the operator with the evidence), or a cheaper certifier that can afford more futures. Each of these is a new hypothesis that needs its own pre-registered evaluation; we did not re-run the evaluation with other parameters.

9 Limitations

Hazards, closures, walk-ins and travel noise are simulated; only the road network is real, and it is a small extract (4 zones, 5 shelters). Futures are generated with the simulator's own dynamics (from observations, not its latent state), so the model class is right by construction; with a real forecast the certificate is only as good as the forecast's calibration (the certifier includes a calibration monitor, unused here). The evaluated systems run without an operator, while the console requires one. Results come from one parameter set ($\varepsilon_{\mathrm{d}} = 0.05$, $\delta = 0.01$, up to 1,024 futures). The pre-registration was internal, not public. Latency targets were missed on a laptop, and no field or bench hardware data exist. The claim-support evaluation behind T2 used construction-labelled pairs written with LLM assistance after an earlier evaluation had failed; they are not human gold labels.

10 Ethics Statement

No human subjects or personal data were involved; the road data are open data. The guideline corpus is public; the LLM verifier received only synthetic claims and public guideline text, under a zero-data-retention setting. The main risk of this line of work is misplaced trust: a "certified" label can make a dispatch look safer than the world model behind it warrants, and a fail-closed gate can *delay* evacuation, shifting risk onto people who are not moved. We therefore report the outcome failure prominently, show failure rates only next to people delivered, keep a human operator in the loop in the console, and refuse and log overrides to uncertified actions—a design choice that itself needs scrutiny by emergency managers before any deployment, since operators may know things the model does not. Nothing here has been used in a real emergency.

11 Conclusion

We built a fail-closed gate that certifies arrival-time feasibility of each evacuation dispatch over keyed futures, commits it atomically with a replayable certificate, and re-certifies groups on the road. Under a pre-registered protocol the gate made no false accept in 400 oracle-checked states and the joint arrival-time indicator was less wasteful than marginal rules, but the gate delivered $21.4\,\%$ of the people a lookahead planner delivered, so it does not improve evacuation outcomes as configured. For deployable decision support we read this as a caution: per-decision safety guarantees should be evaluated, and tuned, against what the system is for.

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A Per-Regime Results

System	Regime	Groups	Safe arr./ep.	Fail. rate	Migr.
HOLD	medium	0	0.0	–	0
HOLD	high	0	0.0	–	0
HOLD	stress	0	0.0	–	0
A	medium	1,045	252.6	18.0 %	0
A	high	946	203.6	29.4 %	0
A	stress	944	213.7	26.1 %	0
A2	medium	1,031	252.0	18.5 %	0
A2	high	960	195.5	33.5 %	0
A2	stress	962	208.2	29.3 %	0
B	medium	620	167.3	4.8 %	0
B	high	362	98.4	3.4 %	0
B	stress	504	138.6	3.6 %	0
P	medium	231	52.5	1.1 %	2
P	high	168	43.2	0.4 %	0
P	stress	259	68.3	0.6 %	0
P-noFC	medium	516	134.4	4.1 %	0
P-noFC	high	275	74.5	2.5 %	1
P-noFC	stress	428	117.6	1.6 %	0
P-noHyst	medium	231	52.5	1.1 %	2
P-noHyst	high	168	43.2	0.4 %	0
P-noHyst	stress	259	68.3	0.6 %	0

Table 3: All systems and regimes, 60 episodes each. Groups: dispatched groups in total; Safe arr./ep.: mean people safely arrived per episode; Fail. rate: mean per-episode group failure rate; Migr.: migrations in total. The stress regime is secondary and was not used for any bar.

B Reproducibility

Every number in this paper is generated from frozen result files by one script, which also writes the tables and figures and a trace from each number to its file and field. Evaluation episodes are seeded and keyed; the analysis script reproduced its output byte-identically on re-run, and an independent machine reproduced a subset of baseline episodes (identical except wall time). The mean P episode took $2{,}251$ s of wall time in the budget run.